

EE5178 - Modern Computer Vision Course Competition

MONOCULAR DEPTH MAP PREDICTION FROM SPAD IMAGES

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ARCHITECTURES TRIED

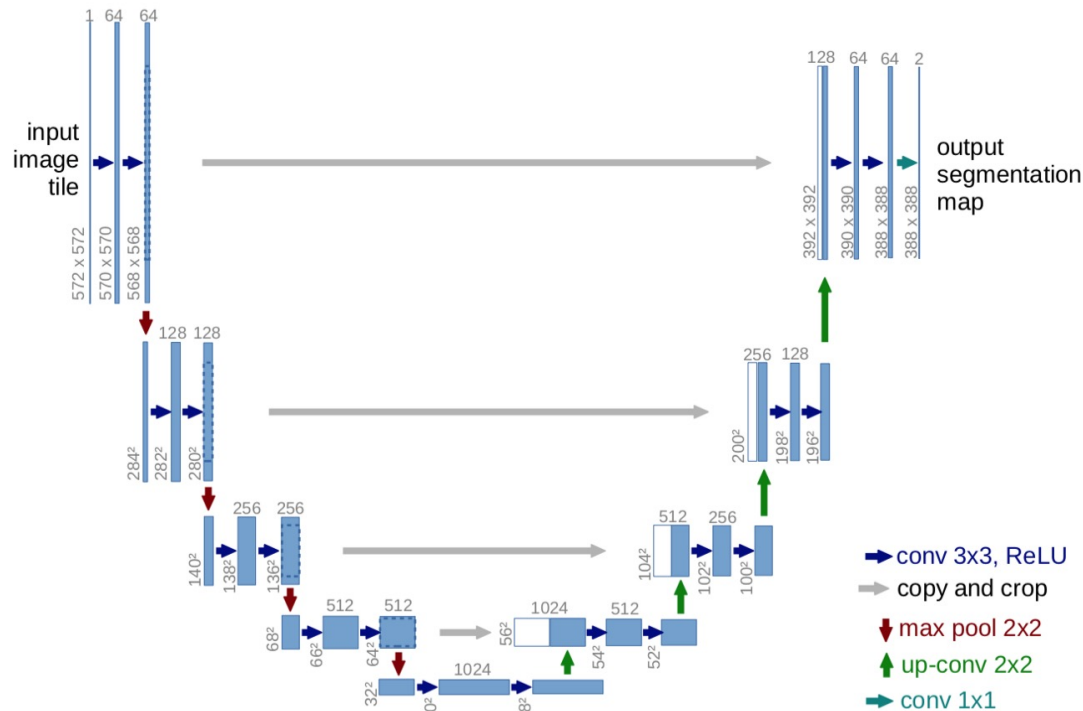


Image Credits : <https://arxiv.org/pdf/1505.04597>

Tried 2 architectures with U-net backbone :

1. Standard U – Net architecture. (MSELoss)

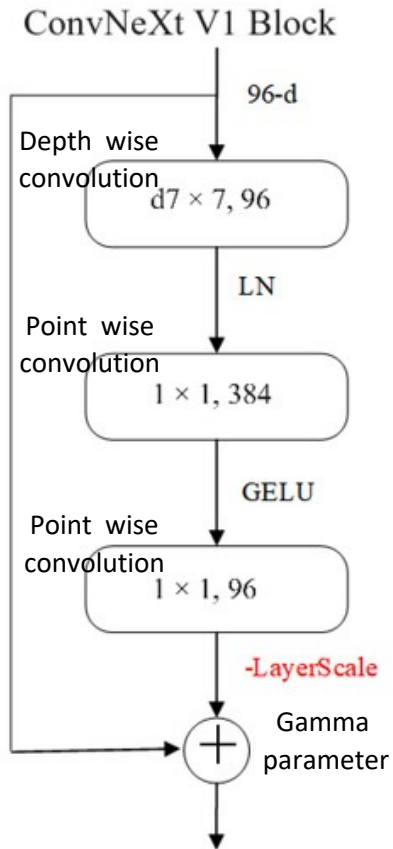
- Base - 16, Bottleneck - 256, Val RMSE - 0.3263
- Base - 32, Bottleneck - 512, Val RMSE - 0.3145
- Base - 62, Bottleneck - 1024, Val RMSE - 0.3026
 - ~ 31.05 M trainable parameters
- Tried MSELoss and L1Loss as training cost function.

2. Custom architecture :

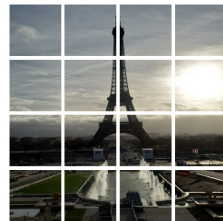
- U – Net Based (encoder – decoder network)
- ConvNext encoder block
- Symmetric ConvNext Decoder block
- Deep supervision through side outputs.
- Fused depth map prediction
- MSELoss as cost function
- Val RMSE - 0.2294
- ~ 33.40 M trainable parameters

BEST ATTEMPT

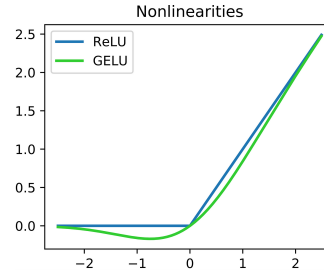
(ConvNeXt Encoder-Decoder Architecture with Multi-Scale Fusion)



Credits : [huggingface](https://huggingface.co)

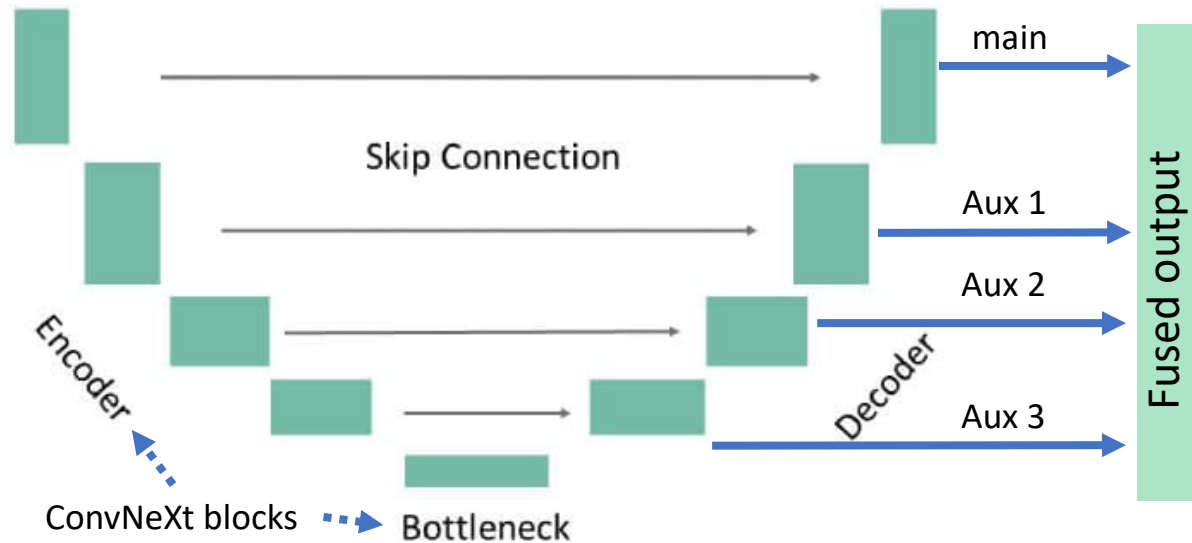
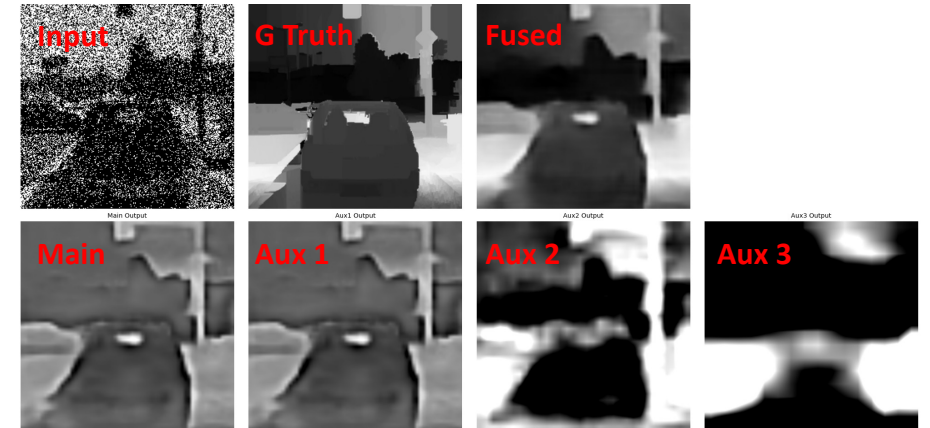


Stem : Patchify
 $4 \times 4, S = 4$



Gaussian Error Linear Unit (GELU):

- Smooth,
- Information-preserving, and
- Boosts model performance.
- Preferred in modern transformers.



Decoder Block :

- Symmetric with encoder.
- Uses depth & point convolutions
- BatchNorm, Relu
- Attention Mechanism (SE) block.
- Refer to the code.

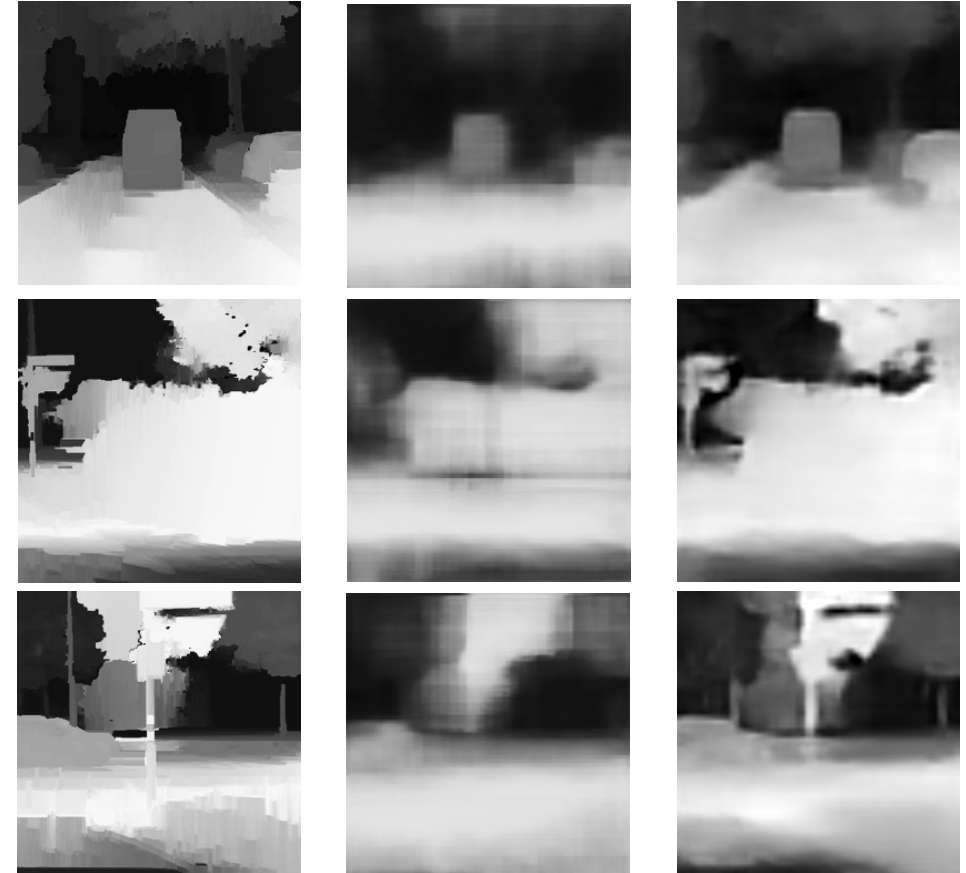
QUALITATIVE RESULTS

Ground truth U – NET (Std) Custom Arch.



Both models didn't capture the subtle depth changes.

Ground truth U – NET (Std) Custom Arch.



Custom Arch. Performed better in capturing **high contrast**.

I wanted to try...

- High contrast depth changes was predicted better. (Deep supervision, fused side outputs)
- To focus on capturing proper **shapes, forms, edges** and retain existing predictability.
- To keep the total trainable parameters less (25 M – 30 M), (MobileNet, ConvNeXt like encoders.)
- **Multi loss cost function strategy (+) loss rebalancing techniques (+) initial patchifying at stem.**
 - [A ConvNet for the 2020s](#)
 - [Multi-Loss Rebalancing Algorithm for Monocular Depth Estimation](#)
 - [Depth Estimation using Weighted-loss and Transfer Learning](#)
 - [SoftAdapt: Techniques for Adaptive Loss Weighting of NN with Multi-Part Loss Functions](#)

Public Leaderboard @ the end : 39/47

Private Leaderboard @ the end : 39/47